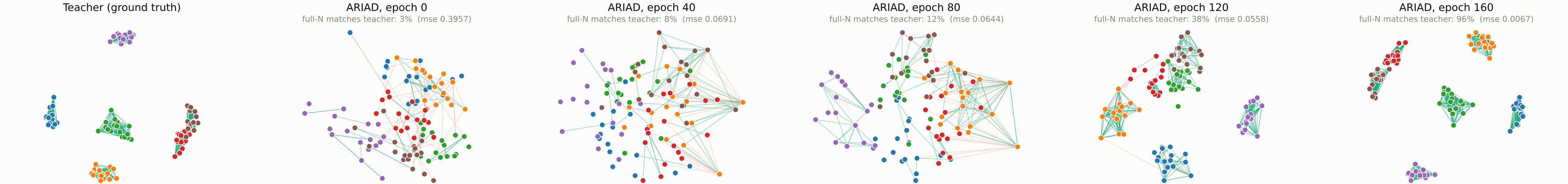


ARIAD's maintained k-NN graph converges to the teacher's (trained at N=2,006, k=16, sparse teacher; 102 pts / 6 clusters shown)



Training uses the FULL N=2,006 tokens (real single-space ARIAD recipe, unmodified) -- this session found recall plateaus well below the real experiments' level when N is only ~100, regardless of embedding dimension, since W_z/W_v are shared across all rows and a tiny N gives a noisy shared-parameter gradient. Only 6 whole clusters (102 of 2,006 points) are drawn for legibility; edges leaving this subset are not rendered, but the annotated recall/mse are the FULL-dataset values. Each edge is one directed ARIAD neighbor link: green if also one of the teacher's true top-16 neighbors, red if not. Model, search and training loop are exactly `ariad_train_single.py`'s; the student's actual embedding is 32-D, PCA-projected to 2D (fit on the drawn subset only, independently per panel -- absolute orientation differs panel to panel; only the graph -- point colour + edge colour -- is the signal to read).